```
interface Lebowski
{
    public string $status { get; }
    public string $beverage  { get; set; }
}
```

Here we've defined `$status` with a `get` property hook and `$beverage` with both `get` and `set`. The implementing class would then have to supply some code for both of those, which fire every time the property is accessed, and I'll show you how to do this later. Note: it is not possible to define a default value for a hooked property in an interface; only the implementing class can do that.

Inheritance and Abstraction

Another fundamental feature of object orientation is *inheritance*. Inheritance is the mechanism by which a class may extend the functionality of another, which facilitates reusability. The inheriting class is known as the *parent* class or *superclass*, the derived class is the *child* or *subclass* (the parent/child relationship giving the mechanism its name).

Parent classes are usually more generalized versions of subclasses. Imagine we had to represent cars in our system, we might have to deal with many different brands and could have classes such as `Volkswagen`, `Ford`, `Audi`, and so on. There will likely be common properties and methods for all of them, and reproducing those common members violates the DRY principle (Don't Repeat Yourself).

If we put those shared members into a parent class of `Car`, then we can use inheritance to ensure that the child classes have the properties and methods they all need, and it only needs to be declared once. The keywords `public`, `protected`, and `private` have a role to play here. Subclasses will inherit `public` and `protected` members, but not `private`. Make sure to use `protected` if you intend for subclasses to inherit a hidden member. Subclasses are also allowed to change the visibility of members, but only to a less restricted level. Thus, `protected` members can become `public`, but not vice versa.

Duplicated code means extra maintenance effort and more chances for bugs to creep in, but by using a parent class (with `public` or `protected` members), we only need to write code once, and any inheriting class automatically has it too.